Suraj Khot

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.42
Sanjay Ghodawat Olympiad School, Ankali	Class XII (Maharashtra State Board)	2024	88.00%
Tatyasaheb Musale Vidyalaya, Ichalkaranji	Class X (Maharashtra State Board)	2022	95.60%

SCHOLASTIC ACHIEVEMENTS

Expert on Codeforces (Max rating: **1630**) Profile

SKILLS

Programming & Scripting Languages: C, C++, Python

Tools & Libraries: Pandas, NumPy, SciPy, Matplotlib, LaTeX

PROJECTS

QUANTITATIVE ALGORITHMIC BACKTESTING FRAMEWORK | C++, Python

- **Engineered** an **event-driven C++ backtesting engine** simulating realistic broker constraints (slippage, commissions, next-bar-open fills) to **structurally eliminate the look-ahead bias** inherent in naive sequential processing.
- Implemented quantitative trading strategies (SMA, Bollinger Bands) and a **drawdown-gated risk manager**, verifying Sharpe, CAGR, and max-drawdown analytics against independent reference models to ensure logical correctness.
- Integrated C++ execution speed with Python analytics via pybind11 and **zero-copy NumPy data transfer**, facilitating a dual-pipeline cross-validation that successfully identified and resolved a **silent floating-point truncation bug**.
- Developed a multi-threaded asynchronous JSON logger to persist trade data without blocking the main execution loop, achieving **zero data loss under concurrent load** and enabling pandas/matplotlib equity curve visualization.

MULTI-PATTERN STRING MATCHING USING AHO-CORASICK AUTOMATON | C++

- Engineered an Aho-Corasick automaton from scratch featuring failure-link and output-link traversal in **O(N+L+K) time** to overcome the scaling limitations of naive multi-pattern matching on massive datasets.
- Reduced search time for 1,000 patterns against the 3.2-billion-character human genome from **~40 days to ~6 minutes (a ~9,600x speedup)**, validating the complexity tradeoff against a suffix-tree alternative.
- Benchmarked algorithmic performance across varying sequence scales to identify the practical efficiency crossover point, demonstrating an **8x speedup** on sequences as small as 2,261 characters and confirming predicted asymptotic behavior.

APPLIED STATISTICS AND ANALYSIS | Python | Course Project (Prof. Arunabha Majumdar)

- Constructed frequency tables and statistical visualizations across discrete and categorical datasets to summarize raw data, successfully identifying right-skewed distributions, near-Normal geometries, and statistical outliers using IQR box-plot rules.
- Computed **Z-based confidence intervals** to quantify point-estimate uncertainty and empirically validated the **Central Limit Theorem** by simulating 1,000 sampling iterations from an exponential population to confirm Normal convergence.
- Programmed one-sample t-tests from first principles for directional (left/right-tailed) and two-tailed scenarios, cross-validating rejection regions against scipy.stats to correctly evaluate null hypotheses at a **0.05 significance level**.
- Executed chi-square tests of independence for categorical associations and manually derived Ordinary Least Squares (OLS) regression coefficients via normal equations, verifying mathematical accuracy against established SciPy libraries.

RELEVANT COURSES

Computer Science: Data Structures and Applications, Data Structures and Applications Lab, Database Management Systems I, Algorithms, Theory of Computation, Compilers I, Introduction to Programming

Mathematics: Applied Statistics, Introduction to Number Theory, Probability Theory, Ordinary Differential Equations, Linear Algebra, Transform Techniques, Convex Optimisation

POSITION OF RESPONSIBILITY

Associate - Finance Domain, Finance and Consulting Club, IIT Hyderabad

Associate - Operations Domain, Entrepreneurship Cell (E-Cell), IIT Hyderabad

Active Volunteer, National Service Scheme.

EXTRACURRICULAR

Secured a **Silver Medal** in the Maharashtra **State-level** Rowing Competition.

Achieved a **Grade A** in the Basic Drawing & Painting Course at Shivaji University.